

Mixed Design in Reality: How to translate a VR Game to a MR World and What's some design issues about it?

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ABSTRACT

Adapting a VR game to MR posed challenges in 360° gameplay, UI, security, and thematic consistency. By prioritizing safety, readability, and engagement, we created an enjoyable experience. In this abstract, we explore the design challenges and decisions taken to address these issues.

CCS CONCEPTS

• **Applied computing** → **Media arts**; • **Human-centered computing** → *User centered design*.

KEYWORDS

Mixed reality, virtual reality, immersive experiences, game design.

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1 INTRODUCTION

Yuki MR is a new game version using Mixed Reality (MR), originally in Virtual Reality (VR) named *Yuki Home*, both being different representations of *Immersive Virtual Environments* (IVEs) [Slater and Usoh 1993a]. As suggested [Slater and Usoh 1993b], some factors may contribute to the participant's sense of presence in such environments, which could be think as external and internal (or subjective) factors. Both of these were considered when we created the MR version. *External factors* are those wholly determined by the hardware and software technology employed to generate the environment, and *internal factors* could be think as how sensory inputs are comprehended by playing the game.

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1.1 Translating the VR to MR

This game combines bullet-hell gameplay with rogue-lite elements, where players assume the role of a child playing with their favorite action figure named Yuki, being assisted by a mentor archetype character named POD, in order to save the Universe from the evil enemies named Yokaliens. Players control Yuki with their dominant hand and POD with the other one.

We translated the original VR game to a MR game following the ideas of presence, immersive storytelling and experiential design proposed by [Bye 2022], introducing a 360° spaciality of real world adding new actions, enemies, bullets, and movements which brought significant challenges to the game design in spatial representation, interfaces and personal safety. Our goal was to maintain the narrative with the original game due to the fact we already have a large and established base of fans with expectations.

The original Yuki game had a rail **gameplay structure**, where enemies came from the infinite horizon against the player, and we had to adjust the gameplay for the new confined space experience. To do this, we added an Arcade genre layer and made sure it was compatible with the new ways of spatial perception and movement exploration, driven by the limitations of the players' physical space.

In addition, we had to redesign the **user interface**, ensuring that players could easily understand the information on the screen while moving around in physical space. Readability and understanding were fundamental to the success of the game and, therefore, we invested in an intuitive and clean interface.

Player safety was a crucial challenge we addressed of. Our user-centered design prioritized safe gameplay that did not jeopardize physical security by considering enemy positioning, fair 360° exploration, challenge construction, and bullet/enemy movements.

In keeping with the game's core fantasy, we kept the original theme and adapted game elements to fit the new reimagining. This allowed players to immerse themselves in the game's story, identify elements they were already familiar with, and explore the world in new and exciting ways.

2 DESIGN DIRECTIONS

In order to have a world representation with virtual elements, the headset must allow to map real elements as walls, doors, windows, and furniture, here called **mapped world**.

Initially, a key challenge was introducing VR and MR concepts to the professionals involved, who were unfamiliar with these technologies. This involved capturing the essence of presence and aligning objectives with the original game. While a preliminary AR

version provided a starting point, significant design changes were necessary to incorporate MR interactions.

We adopted a wave-based approach in the transition from VR to MR, gradually introducing MR elements to prevent user disorientation. This involved incorporating three wave types: *ordinary waves* visible only in MR, and two hybrids VR-MR visualization, a *bonus waves* (including flocking and lasers), and *boss waves*, featuring bosses. These goals guided our design decisions:

- The MR experience.
- A comfortable experience.
- Maintain the core concept and fantasy.

2.1 The MR Experience

The most important goal is the MR representation which presupposes not just display virtual elements over real world on the headset, but try to mix it with the mapped world, creating unexpected but interesting interactions. The user's feeling should be the "reality is broken" and we create virtual elements to reach it, such as these:

- **POD Punch:** In the original version, POD assisted Yuki in collecting rewards, providing narrative context and hints. In MR mode tests, players wanted to use POD for close-range combat. This addition greatly enhances the MR experience, as POD can punch enemies, causing collisions, and bounces.
- **Portals:** In the MR version, enemies are spawned through virtual portals that consider the mapped world, adjusting shape, scaling, and positioning relative to real elements.
- **Collisions:** All game elements, such as enemies, bullets, and the POD Punch action, interact with the mapped world. Enemies navigate around the scenario, with bullets bouncing off surfaces. Yuki can activate a temporary shield that protects her from hits and causes bullets to rebound.
- **Bullet formations:** All the bullet formations were refactored to create a closer suitable version for the new 360° necessities. Some formations create an *area denial space* forcing players to dodge and move their bodies through the space.
- **Enemy movements:** Enemies in the original game just moved through a rail trail against Yuki and POD, performing some circular movements. For MR mode, enemies must fill the entire space around the player exploring the mapped world.
- **Flocking and Lasers:** We added two new enemy types to the game. The first is a *flocking enemy* that flies throughout the space, creating an immersive MR experience. The second type is a *wall-mounted enemy* that casts lasers, creating an area denial space. Flocking enemies also have the ability to drop bombs, further enhancing the area denial aspect.
- **Enemy shield:** Another addition was enemies with shields which can be broken using POD Punch action, bouncing against other enemies and the mapped world.
- **Collectibles:** Another element that promotes player movement is the presence of stationary collectibles (or power-ups) that are dropped by enemies when they are killed.
- **Radar:** We added a 360° radar displaying enemy positions. The radar features a central arrow representing the player's head position and direction. Enemies are indicated by three icons: an upward arrow for above, a downward arrow for below, and a square for enemies at the same height.

2.2 A Comfortable Experience

As we are creating a mixed experience, we need to avoid an unfair gameplay — for instance, enemies coming through behind the player and shooting on them. To mitigate it, the **Yuki's shield** was changed from the original front-face shape, suitable for a rail gameplay, to a 360° shape. Bullets coming from any directions can be avoided using it.

The **head comfort zone**, which refers to the frontal area where the player's attention is focused, should be taken into consideration throughout the entire experience. This doesn't mean that we can't utilize the 360° space, but it is important for the user to be aware of where the action is happening, even if we shift their focus. To address this, portals were strategically placed in a clockwise and counter-clockwise direction, following a smooth pacing challenge. This ensures that a portal never appears behind the player without prior communication. Additionally, the implementation of **spatial sound** enhances players' directional perception, providing them with a better sense of the game's audio cues.

For last but not least, the **transition between VR and MR** environments was created with a huge portal increasing slowly coming from the ceiling and stopping around a 50% of the entire real environment for the player safety and preventing user desorientation.

2.3 Maintain the Core Concept and Fantasy

The captivating aesthetic and narrative are key elements in Yuki's game. Therefore, it is essential to prioritize the beauty of the enemies' behavior, the formation of bullets, and maintain a seamless gameplay experience as guiding principles for both the gameplay and the UI/UX design. Additionally, it is crucial to always keep in mind the central concept of the player being a child playing with an action figure. For instance, it would be ill-advised to incorporate a virtual arm representing the child grabbing Yuki, as it contradicts this concept and may disrupt the immersion and overall experience.

Considering this, we maintain the core concept and phantasy reusing and inheriting all possible content adapted for the MR experience. Some **enemies and bullets designs** were changed to have a better contrast against the background of real world, which can be complex in texture patterns.

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